Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/19/2005
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Project Title:	Evaluation of the Late Effects of Heavy-Ion Radiation on Mesenchymal Stem Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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	Differentiation of pluripotent stem cells is the complex multi-step process by which the genetic program designated for a specific cell lineage by a complex biological process is stably expressed. This includes the expression of lineage-specific genes that mediate the associated biological functions and restriction of the proliferative capacity irreversibly during terminal differentiation. Defects in the differentiation of mesenchymal stem cells have been shown to play an important role in many forms of cancer. Moreover, de-differentiation of epithelial stem cells to a mesenchymal phenotype has also been shown to be a hallmark to metastatic cells. However, the initiation event, which eventually leads to carcinogenesis, has been elusive. It has been postulated that double stranded DNA breaks, which have been shown to be the main type of DNA damage caused by HZE radiation, could cause both DNA mutagenesis and, with proliferation, genomic instability. The goal of this proposal is to answer the following question: Does differentiation of pluripotent stem cells play a role in		

Task Description:	initiation, promotion, and/or progression of genotoxic damage caused be HZE radiation ultimately leading to carcinogenesis? The effect of HZE radiation on human derived mesenchymal stem cell (MSC) differentiation is tested via three specific aims. (1) Compare the initial effects of HZE and gamma radiation on MSC chromosomal aberrations. This aim will be tested by exposing mesenchymal stem cell in either proliferative or differentiating states and examining chromosomal aberrations. (2) Evaluate the effect time of culture after exposure to HZE radiation on MSC's function. Both chromosomal aberrations and gene expression will be analyzed to examine the effect of differentiation on the promotion-step of the model. (3) Compare the effect of localized microenvironment of the culture system and the LET of the charged particles on MSC differentiation and carcinogenic potential. Tissue-engineering principles will be utilized to construct functional 3-dimensional tissue-like constructs, which then can be used as a model to test the progression of carcinogenesis. The biological endpoints tested in this aim will be gene expression and histological and immunohistochemical evaluation during extended cell-culture. In each of the specific aims, a unifying analysis scheme is included to easily evaluate the effect of differentiation on the three steps leading to carcinogenesis. The completion of these studies will represent a significant contribution to the understanding of the role of differentiation in genotoxic damage induced by HZE radiation. In addition, these studies may potentially lead to the use of this human in vitro model for the development of strategies to minimize or eliminate these harmful events.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Please note that this is a new grant in FY 2004. There is no research progress reported for this period. For more information on this grant or this PI, please contact the Task Book Help Desk at taskbook@nasaprs.com.
Bibliography Type:	Description: (Last Updated:)